
A Machine Learning Analysis of Calibration on Manifold Markets

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Abstract

Prediction markets produce probability estimates by aggregating crowd beliefs, but their reliability varies across market conditions. This paper investigates when and why prediction market prices are miscalibrated, using Manifold Markets, a large play-money platform with a public bulk data export. We construct a dataset of 25,275 resolved binary markets, each observed seven days before resolution, and train five machine learning models against the raw market price. No model beats the crowd globally, confirming that the crowd is well-calibrated on average. Subgroup analysis reveals structural conditions where the crowd is miscalibrated; in one case, crypto-speculation, the Random Forest model improves on the crowd by exploiting systematic directional underpricing.

1 Introduction

Prediction markets are platforms where participants buy and sell contracts on future outcomes of real-world events. The price of a contract reflects how likely the crowd collectively believes that the event will occur. In other words, the price aggregates information from many participants into a single probability estimate, and prior research indicate that these are strong forecasting mechanisms. [1, 2, 4].

However, the quality of the crowd’s estimate is not uniform. For example, a market pricing a crypto event is not the same as one pricing a well-understood political outcome. Prices should be more reliable when more information has been aggregated, but characterizing the conditions that cause miscalibration requires more analysis.

Our research question is: **can machine learning identify the structural conditions under which a prediction market price is a poor probability estimate?** We study this on Manifold Markets, a play-money platform with a large public dataset spanning 130,091 markets across 22 topic categories from 2022 to 2024. Because Manifold is play-money, the absence of financial incentives might augment the miscalibration patterns that real-money platforms like Polymarket might not have.

2 Related Work

Prediction markets are well-calibrated, but not uniformly. [2] find that markets are well-calibrated close to resolution but show a systematic favorite-longshot bias at longer time horizons. This means that low-probability events are overpriced, while high-probability events are underpriced, with the bias growing as time to resolution increases. Our subgroup analysis tests whether this holds true in Manifold markets. Evidence on the longshot bias is mixed across platforms: Bürgi, Deng, and Whelan [7] document it on Kalshi, while [4] find no such bias on Polymarket.

Machine learning augmentation of collective forecasts. [3] use a 43-feature random forest model to identify higher-accuracy forecasters and construct human-machine hybrid forecasts that outperform

the market forecast when the two disagree. While their model reweights individual forecaster input, our model operates at the market level: we use market structure, participation, category, and price-dynamics features to identify conditions under which the already-aggregated crowd price is likely to be miscalibrated. Our evaluation also follows work on probabilistic classification. [8] show that common supervised classifiers differ substantially in the quality of their predicted probabilities, even when their classification accuracy is similar. Our models are trained as binary classifiers, but evaluated as probability estimators using Brier score against the crowd-price baseline.

Structural drivers of miscalibration. [5] find that roughly 3% of Polymarket accounts are classified as skilled and drive much of price discovery, while most other accounts do not improve accuracy. This motivates our trader-tier analysis. Since we don’t have trader specific information, we use market-level proxies (trader count, volume, bet frequency) as stand-ins for informed trader presence.

[6] decompose calibration variance across Kalshi and Polymarket into a universal horizon effect, domain-specific biases, and domain-by-horizon interactions. We study the domain-specific biases in our analysis as well. Our Manifold results show that politics-related categories are best-calibrated (politics Brier 0.0698), while sports are among the weakest (nfl Brier 0.1417). This suggests that there are domain-specific calibration patterns, and it depend on platform incentives and trader composition.

Where our work fits in. We extend the work to an ML-based miscalibration analysis to Manifold Markets. Our contributions are: (1) a clean 25,275-market dataset with 43 features per market, snapshotted consistently at seven days before resolution, with a chronological train/test split to avoid data leakage; (2) a comparison of five ML models against a crowd price baseline across global and subgroup evaluations; (3) identification of market conditions where crowd price is least reliable; and (4) the finding that ML generally fails to correct those errors globally, with one exception in the crypto-speculation category.

3 Data

3.1 Manifold Markets

In Manifold Markets, users receive a virtual currency. Manifold provides a public bulk data export that includes all markets, bet histories, and comments. We use the July 2024 dataset, which contains `manifold-contracts-20240706.json` (130,091 total markets) and `bets.json` (6.3 GB of full bet histories). We do not use the comments dataset due to high sparsity.

3.2 Preparing the Data

Starting from the raw markets, we apply the following filters: binary resolution only (YES/NO outcomes); resolved YES or NO (excluding N/A and cancelled); at least 10 unique traders; at least 100 mana in trading volume; and duration longer than 7 days. These filters yield 25,401 markets, of which 126 are dropped due to missing snapshot features, leaving a final dataset of **25,275 markets**.

For each retained market, we compute all features at a fixed point: **seven days before the market’s resolution time** (`resolutionTime - 7 days`). This ensures that every feature reflects information strictly available before resolution, preventing data leakage. The seven-day window provides a sufficient observation point where the market price has had time to converge, while maintaining a consistent structure across markets of different durations.

We use a time-based split: markets resolving before February 2024 form the training set (20,220 markets), and markets resolving February–July 2024 form the test set (5,055 markets). Random splitting is inappropriate here because it allows future market information to leak into training folds. We validated this empirically when random cross-validation during tuning caused RF to perform worse on held-out time periods (Brier 0.1514 with random CV vs. 0.1221 with time-based CV).

The final dataset of 25,275 markets has a class imbalance: **38.3% resolve YES** (9,583 markets) and **61.7% resolve NO** (15,692 markets). All models use `class_weight=‘balanced’` during training. The 43 model features span six groups: market price signals, participation metrics, price dynamics, market structure, question text features, and 22 one-hot-encoded topic category indicators.

Table 1: Dataset summary

Raw markets	130,091
After filtering	25,275
Retention rate	19.4%
YES resolutions	9,583 (38.3%)
NO resolutions	15,692 (61.7%)
Training set (Jan 2022 – Feb 2024)	20,220
Test set (Feb – Jul 2024)	5,055
Features	43 (21 numeric + 22 category one-hot vectors)

Table 2: Full 43 feature set

Feature	Source	Meaning
<i>Price signals (4)</i>		
price_at_snapshot	bets	Crowd probability 7 days before close
initial_prob	market	Creator’s opening probability
price_change_7d	bets	One week price change (momentum)
abs_price_from_half	derived	$ \text{price_at_snapshot} - 0.5 $
<i>Participation (4)</i>		
num_traders_at_snapshot	bets	Unique traders before snapshot
volume_at_snapshot	bets	Total Mana traded before snapshot
bet_count_at_snapshot	bets	Number of bets before snapshot
bet_frequency	derived	$\text{bet_count} / \text{time_elapsed}$ (bets per day)
<i>Dynamics (3)</i>		
price_volatility	bets	Std. dev. of bet probabilities before snapshot
last_bet_gap_days	market	Days between last bet and resolution
volume_last_24h	bets	Mana traded in 24h window before snapshot
<i>Structure (6)</i>		
planned_duration_days	market	$\text{closeTime} - \text{createdTime}$ in days
time_elapsed_at_snapshot	derived	Days from creation to snapshot
market_tier	market	Creator commitment: play=0, basic=1, plus=2
elasticity	market	Price sensitivity to marginal bet
total_liquidity	market	Liquidity pool size (Mana)
total_fees	market	Sum of creator, platform, liquidity fees
<i>Question text (3)</i>		
question_length	derived	Character count of question text
question_has_year	derived	Binary: year mentioned in question
question_has_pct	derived	Binary: percentage mentioned in question
<i>Market flag (1)</i>		
has_category	market	Binary: market has a category tag
<i>Category one-hot (22)</i>		
cat_ai, cat_crypto_speculation, ...	derived	One-hot encoding of topic slug

4 Methods

4.1 Baseline

Our baseline is the raw market price at snapshot time. We use this as the probability the market resolves to YES since it represents the collective belief of the crowd at that point. A model that cannot beat this baseline is not extracting signal beyond what the market already incorporates.

4.2 Models

We selected five models that cover linear, tree-based, neural network, and kernel-based approaches to see which model family performs the best and whether any model family improves on the crowd

calibration. These models were trained on the same 43-feature matrix. All hyperparameters were selected via `TimeSeriesSplit(n_splits=5)` CV, which prevents any data leakage.

- **Logistic Regression** ($C = 0.1$, `StandardScaler`): linear baseline; interpretable via coefficient magnitudes.
- **Random Forest** (500 trees, $\sqrt{\text{max_features}}$, balanced class weights): best-performing model; tree-based splits require no feature scaling.
- **XGBoost** (`max_depth=5`, `lr=0.01`, `subsample=0.7`, `colsample_bytree=0.8`): gradient-boosted trees; `scale_pos_weight` corrects for class imbalance.
- **MLP** ($128 \rightarrow 64 \rightarrow 32$, ReLU, $\alpha = 0.001$): early stopping on a 10% validation hold-out; `StandardScaler` applied.
- **SVM** (RBF kernel, `probability=True`): Platt scaling converts margin scores to calibrated probabilities; `StandardScaler` applied.

4.3 Evaluation

Our models and our baseline are both probabilities, so we use Brier score as a primary metric. Brier score measures probabilistic forecast error, penalizing both miscalibration and overconfident wrong predictions. For binary outcomes with predicted probabilities in $[0, 1]$, the Brier score ranges from 0 to 1. A lower Brier score is better. $BS = \frac{1}{N} \sum_{i=1}^N (f_i - o_i)^2$, where f_i is the predicted probability and o_i is the binary outcome (0 or 1).

5 Results

5.1 Global Model Comparison

Table 3 reports the performance on the 5,055-market test set. No model beats the baseline globally. Random Forest achieves the smallest gap (Brier 0.1221 vs. 0.1071), but still underperforms the crowd. The baseline also scores the highest on AUC-ROC (0.9289), meaning the crowd is able to discriminate between YES and NO outcomes better than any trained model.

Table 3: Model performance on test set (5,055 markets)

Model	Brier	Δ vs. Baseline	AUC-ROC
Baseline (crowd price)	0.1071		0.9289
Random Forest	0.1221	−0.0150	0.9177
Logistic Regression	0.1407	−0.0336	0.9063
MLP	0.1565	−0.0494	0.8888
XGBoost	0.1610	−0.0539	0.8592
SVM	0.1762	−0.0691	0.8266

5.2 Why No Model Beats the Baseline

Feature importance analysis reveals that the RF is learning a simple heuristic rather than complex market behavior. `price_at_snapshot` alone accounts for 41% of all splitting decisions, and the top four features cover 60% of cumulative importance collectively. These are all crowd-signal features, so the model is essentially learning what the crowd’s currently probability estimate is and how it got there using those dominant features. The remaining 39 features, including all 22 category one-hots likely add more noise than signal.

This helps explain the model ranking. RF wins because 500 trees averaging over feature splits produces stable probability estimates without losing the nonlinear patterns that actually matter for the crowd-price signal. For XGBoost, each tree learns from the errors of the last because it fits trees sequentially to residual errors. This can amplify training-period specific patterns. When the distribution shifts, it doesn’t adjust. MLP likely has too much flexibility but not enough signal from our features, so it stays close to the market-price baseline. For SVM, Platt scaling is less reliable near

the extreme market probabilities, so any market price near the extremes will cause SVM to perform worse. Ultimately, every model faces the problem. `price_at_snapshot` already captures most of the predictive signal. The market price already aggregates most of the information, leaving little additional signal for the models to learn.

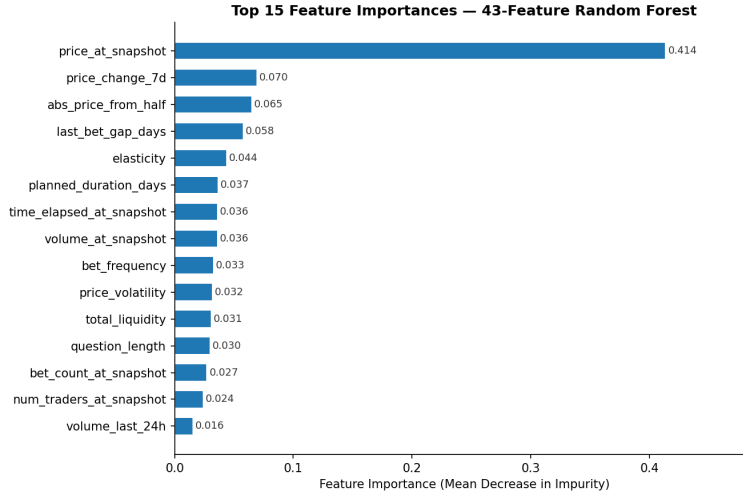


Figure 1: The top four features are crowd-signal features.

5.3 Calibration Plot

Figure 2 shows the crowd price and RF predicted probability against the actual YES rate per price range. The crowd price tracks the perfect-calibration diagonal closely. RF has larger variance from the diagonal in both extreme prices.

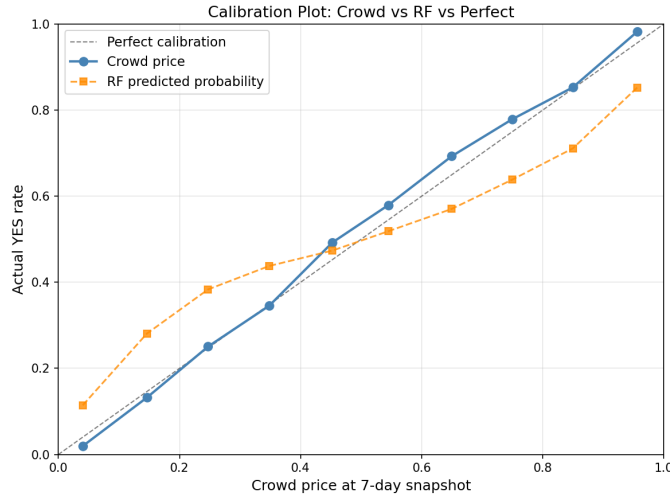


Figure 2: RF regresses toward 0.5 at both extremes.

6 Analysis

We first ask where the crowd price is weakest (baseline Brier per subgroup), and then ask whether RF (our best model) can improve on the crowd within those subgroups. We partition the test set into subgroup slices and compute the baseline Brier score within each slice. Figure 3 shows the subgroups baseline Brier scores, anchored by the global baseline (0.1071).

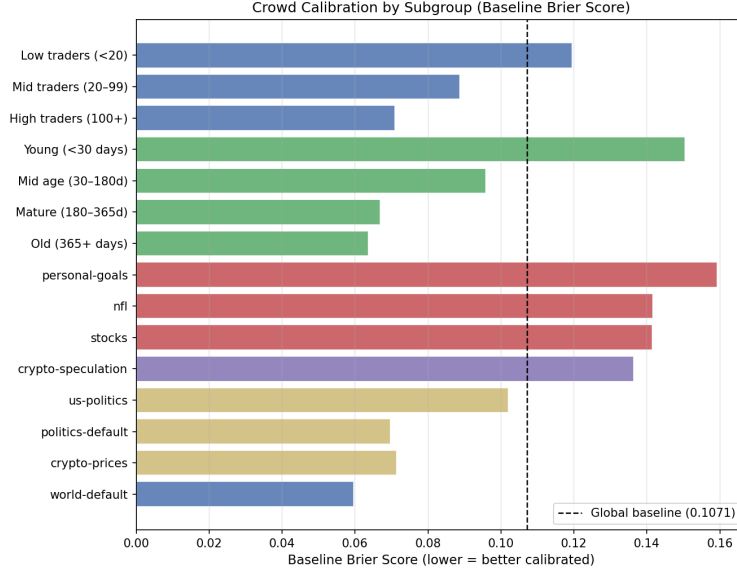


Figure 3: Lower Brier Score is better. Longer bars indicate the least accurate crowds.

6.1 Trader Count Tier

Thin markets are less accurate. Per [5], markets with fewer traders are less likely to contain informed participants who can move the price toward the true probability. The crowd bias, computed by taking the average(predicted probability - actual outcome) is ≈ 0 across all tiers, meaning the crowd’s errors are directionless noise rather than a systematic pattern ML can learn. Thus, RF still underperforms within these subgroups.

Table 4: Trader count tier subgroup results

Tier	N	Crowd Bias	Baseline Brier	RF Brier	RF Δ
Low (<20 traders)	3,098	+0.0004	0.1195	0.1328	-0.0133
Mid (20-99)	1,805	-0.0135	0.0888	0.1056	-0.0169
High (100+)	152	-0.0112	0.0710	0.0986	-0.0276

6.2 Momentum

Momentum measures how much the crowd price moved in the 7 days before the snapshot. A market with strong downward momentum means that traders updated toward NO, while strong upward momentum means traders updated toward YES. We found that markets with strong downward momentum are still overpriced by 8.0 percentage points relative to their actual YES rate, and markets rising sharply are underpriced by 7.4 points. Traders under-adjust to the new information that drives price change. We found that RF struggles to correct this because the correction is subgroup-specific. `price_at_snapshot` feature dominates, so the model mostly learns to track the crowd price rather than adjust for momentum. This suggests that subgroup-specific training or interaction features that amplify momentum effects may be able to exploit this bias more effectively.

Table 5: Momentum subgroup results

Momentum Bin	N	Crowd Bias	Baseline Brier	RF Brier	RF Δ
Strong down (< -0.15)	314	+0.080	0.0896	0.1156	-0.0259
Mild down ($-0.15-0$)	1,337	+0.018	0.0912	0.1101	-0.0190
Stable (0.0)	2,215	-0.011	0.1169	0.1284	-0.0115
Mild up ($0-0.15$)	962	-0.035	0.1154	0.1296	-0.0142
Strong up ($> +0.15$)	227	-0.074	0.0942	0.1075	-0.0133

6.3 Category

Crypto-speculation is the only category where RF clearly beats the crowd ($N=137$, $\text{RF } \Delta = +0.0160$). The crowd prices these markets at 41% on average, but 55% of them actually resolve YES, meaning the crowd systematically underestimates the likelihood of crypto events happening. This consistent 14-point directional error is the one setting where RF learns a useful correction. The features (price volatility, momentum, market age) detects the underpricing and corrects for it. Crypto-speculation is easier for RF to correct because the bias is tied to an explicit category feature. The model likely learns a relatively simple rule that crypto-speculation markets are underpriced on average.

Table 6: Category subgroup results (selected, sorted by baseline Brier)

Category	N	Crowd Bias	Baseline Brier	RF Brier	RF Δ
personal-goals	115	+0.032	0.1593	0.1705	-0.0112
nfl	28	+0.053	0.1417	0.1638	-0.0221
stocks	110	+0.044	0.1415	0.1713	-0.0298
crypto-speculation	137	-0.142	0.1365	0.1205	+0.0160
finance	52	-0.067	0.1277	0.1254	+0.0023
ai	228	+0.003	0.1063	0.1258	-0.0195
us-politics	422	-0.002	0.1020	0.1014	+0.0005
crypto-prices	23	-0.078	0.0714	0.0926	-0.0212
politics-default	381	+0.023	0.0698	0.1063	-0.0366
world-default	76	-0.009	0.0597	0.0766	-0.0169

6.4 Market Age at Snapshot

Calibration improves consistently as markets age, from Brier 0.1504 for markets under 30 days old to 0.0636 for markets open more than a year. This replicates the time-horizon finding of [2], extended here to a play-money platform. RF underperforms across all age bins and worsens for older markets. Crowd price converges to the true probability with time, which explains RF’s worse performance.

Table 7: Market age subgroup results

Age Bin	N	Avg Age (days)	Baseline Brier	RF Brier	RF Δ
Young (<30 days)	1,672	14.5	0.1504	0.1578	-0.0074
Mid (30–180 days)	2,240	87.0	0.0958	0.1133	-0.0175
Mature (180–365 days)	758	255.1	0.0669	0.0861	-0.0192
Old (365+ days)	385	470.7	0.0636	0.0886	-0.0250

7 Conclusion

The Manifold crowd is well-calibrated on average and no ML model improves on it globally. While this was expected, the subgroup analysis succeeds at identifying miscalibration. We showed that RF can fix it in one narrow setting, but better models with better features could outperform within other subgroups as well. This has practical value for researchers using market prices as forecasting mechanisms and for identifying slices where the crowd is consistently underpricing.

We acknowledge the limitations of our study as well: (1) Manifold is play-money, so it might encode signals not present in real-money markets, (2) we have a limited feature set: no text embeddings, news signals, or individual trader histories. We attempted text embeddings but found that 95.5% of markets had no meaningful comment data.

For future work, we want to do a learning-curve analysis to test how the amount of training data or the quality of the features affect the model performance. We also want extend the analysis into real-money markets with a richer feature set. This may give the models additional signals to close the gap or surface more important features other than price_at_snapshot.

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